Mateusz Golczyk

Curriculum Vitae

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Professional Summary

Astronomy graduate with strong background in computational modeling, high−performance computing, and numerical simulations. Experienced in research projects in astrophysics, quantum computing, and detector development at international institutions. Skilled in Python, C++, MATLAB, and parallel computing (OpenMP, MPI). Seeking to apply analytical and programming expertise to engineering and simulation roles in aerospace and defense industry.

Education

2021−2025 **Bachelor's Degree in Science**, Major: Astronomy, *Nicolaus Copernicus University*, Torun, Poland

2017−2020 **Honors High School Diploma**, *Princess Elizabeth General High School*, Szczecinek, Poland

Research Experience

2024−2025 **Research Collaborator**, Florida Institute of Technology, Florida

- Developed quantum computing algorithms for applications in high-energy physics
- Contributed to the testing and development of silicon detectors for the Large Hadron Collider (LHC)

2023−2024 **Research Assistant**, Nicolaus Copernicus University, Poland

- Conducted theoretical development and performed numerical simulations of interacting electron–hole systems within the formalism of second quantization
- Applied advanced methods of computational physics and quantum many−body theory

2024−2024 **Research Assistant**, University of La Laguna, International Internship, Canary Islands, Spain

- Developed and implemented many-body models to study electrical noise in silicon quantum dot networks

- Applied advanced computational physics methods and programming (Python/C++/MATLAB) to simulate quantum dot systems

2022–2024 **Research Project: N–body Simulations**, Nicolaus Copernicus University, Poland

- Performed large–scale N–body simulations of collisional processes in globular clusters.
- Analyzed computational scaling of algorithms ($O(N^2)$ vs $O(N \log N)$) and compared integrators.
- Applied high-performance computing with OpenMP and MPI on multi-core systems to study efficiency and parallel speed-up.

Skills

- **Programming**
  Python, C/C++, MATLAB, Bash, Fortran
- **High–Performance Computing**
  OpenMP, MPI, numerical simulations, parallel computing
- **Data Analysis**
  Statistical analysis, modeling & simulation, machine learning (basics).
- **Tools**
  Windows, Linux, MacOS

Honors & Awards

- Prime Minister's Scholarship 2020, Poland
- Prime Minister's Scholarship 2019, Poland